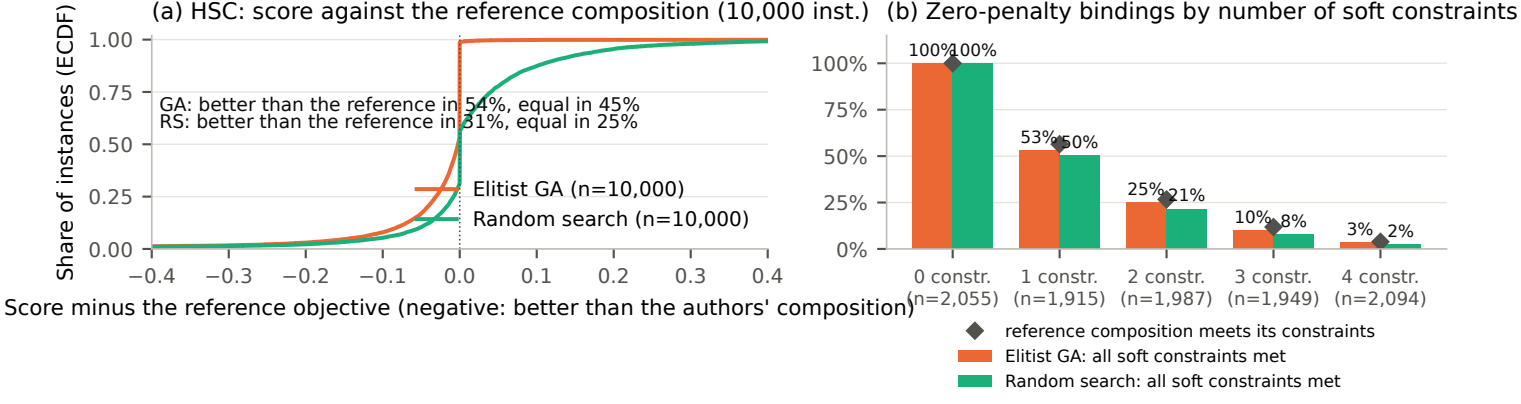
(a) HSC: score against the reference composition (10,000 inst.)
(b) Zero-penalty bindings by number of soft constraints
Share of instances (ECDF)
1.00
0.75
0.50
0.25
0.00
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
GA: better than the reference in 54%, equal in 45%
RS: better than the reference in 31%, equal in 25%
Elitist GA (n=10,000)
Random search (n=10,000)
Score minus the reference objective (negative: better than the authors' composition)
100%
75%
50%
25%
0%
100%
100%
53%
50%
25%
21%
10%
8%
3%
2%
0 constr. (n=2,055)
1 constr. (n=1,915)
2 constr. (n=1,987)
3 constr. (n=1,949)
4 constr. (n=2,094)
reference composition meets its constraints
Elitist GA: all soft constraints met
Random search: all soft constraints met